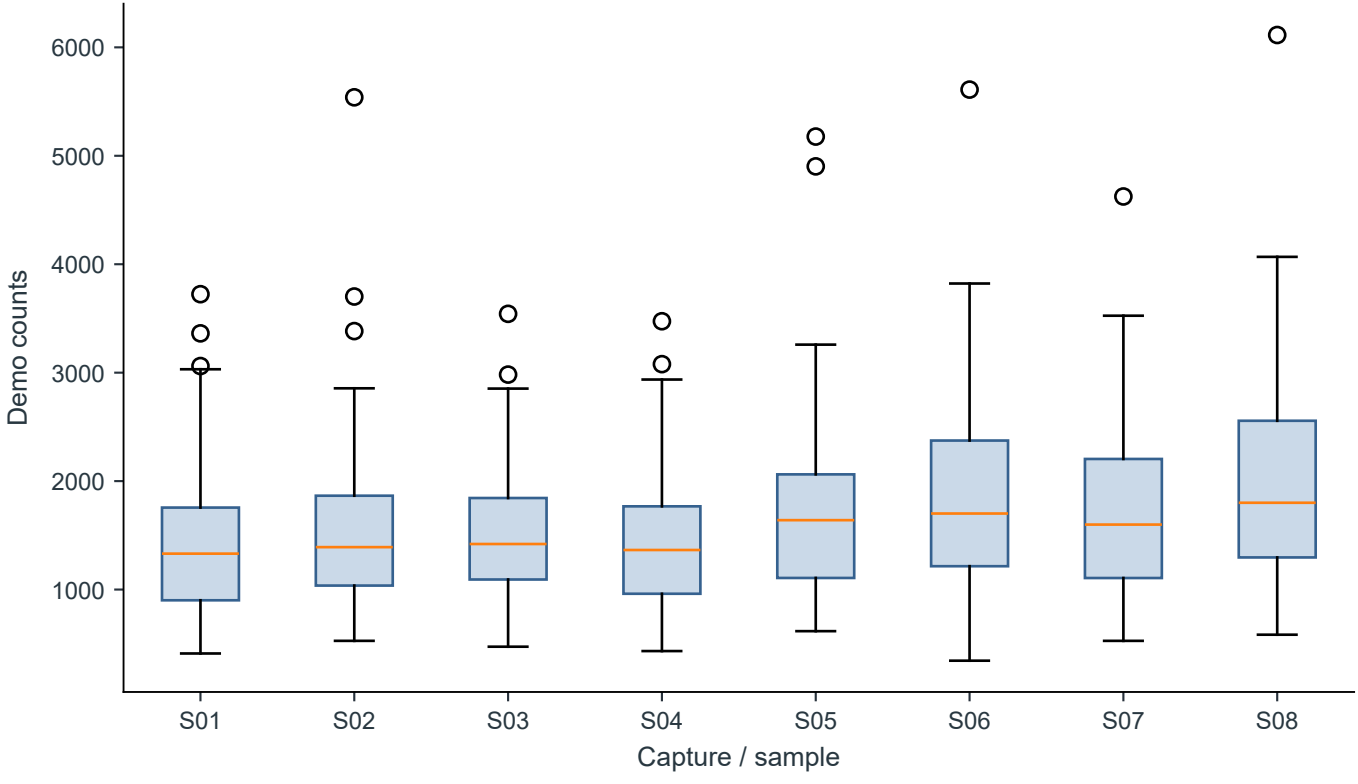Sample quality control